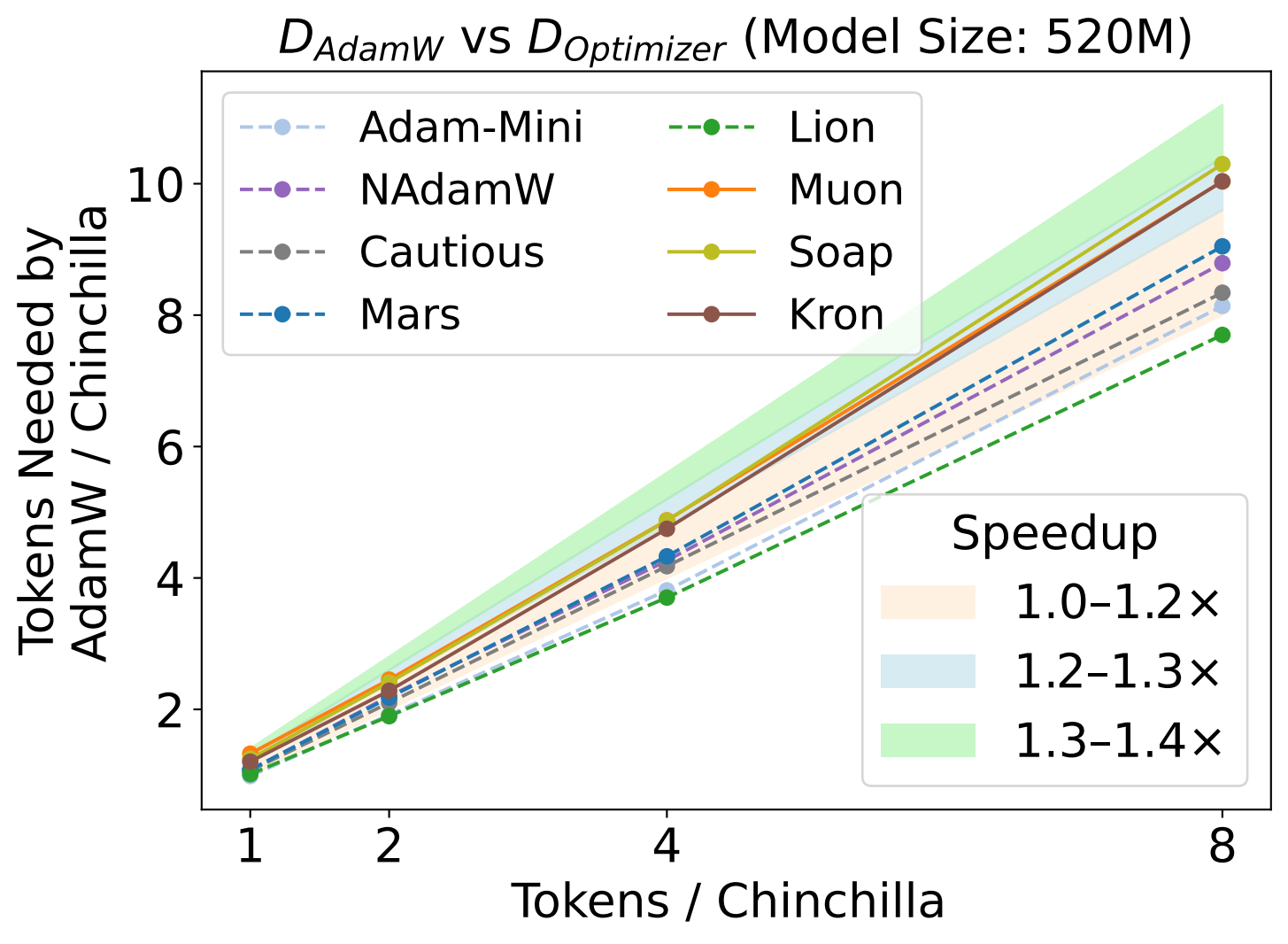
D_{AdamW} vs $D_{Optimizer}$ (Model Size: 520M)
Adam-Mini
NAdamW
Cautious
Mars
Lion
Muon
Soap
Kron
Speedup
1.0–1.2×
1.2–1.3×
1.3–1.4×
Tokens Needed by AdamW / Chinchilla
Tokens / Chinchilla
2
4
6
8
10
1
2
4
8